

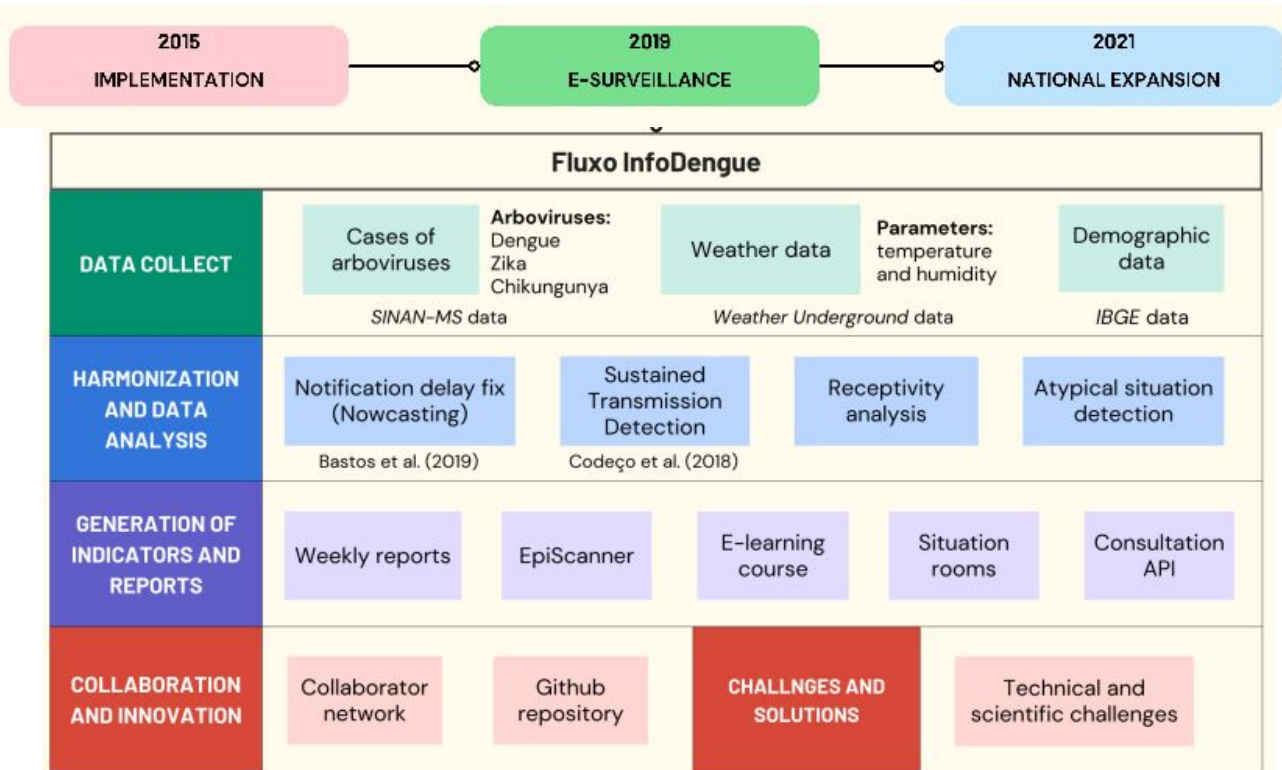
Data integration and predictive ensembles for arbovirus surveillance in Brazil

Luiz Max Carvalho*

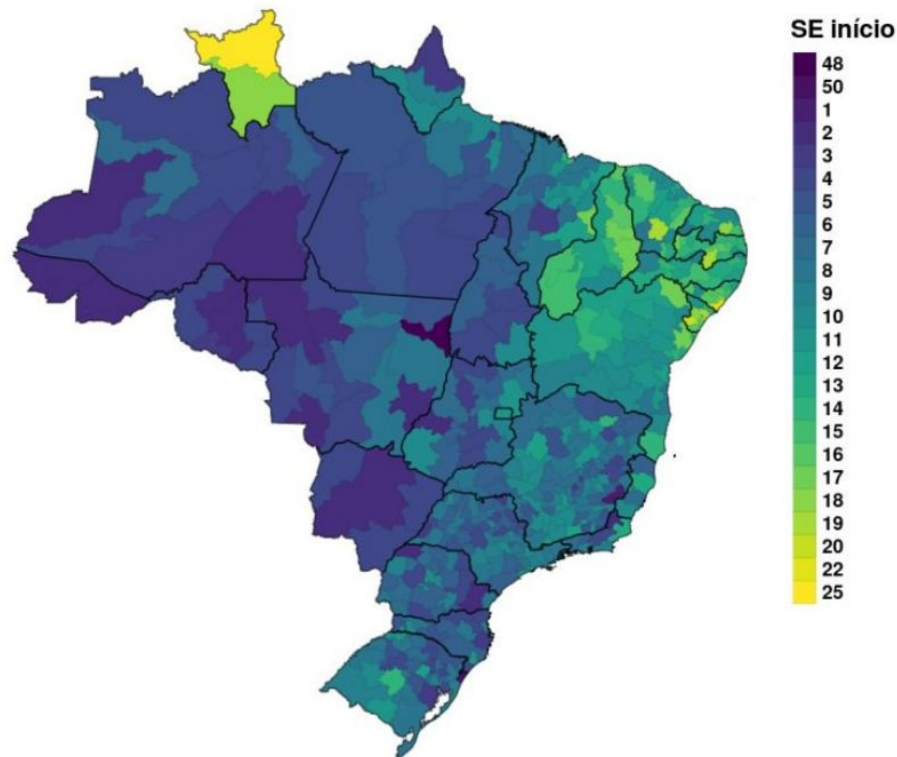
School of Applied Mathematics,
Getulio Vargas Foundation,
Rio de Janeiro



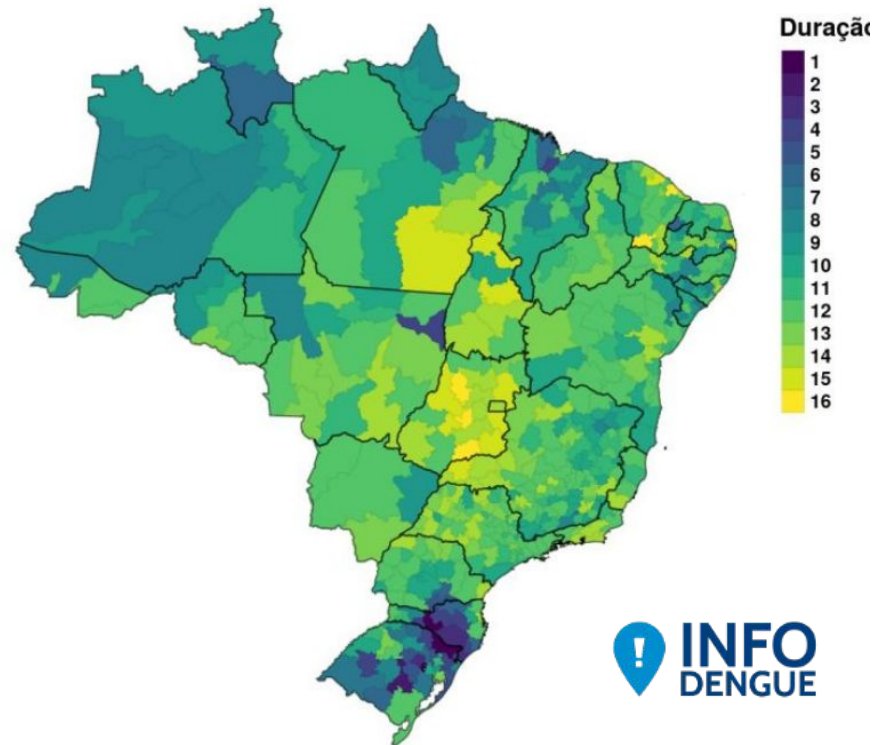
weekly alerts
dengue & chikungunya
5570 municipalities,
27 states



Dengue season onset



Season duration

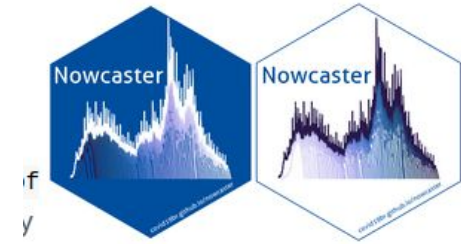


RESEARCH ARTICLE

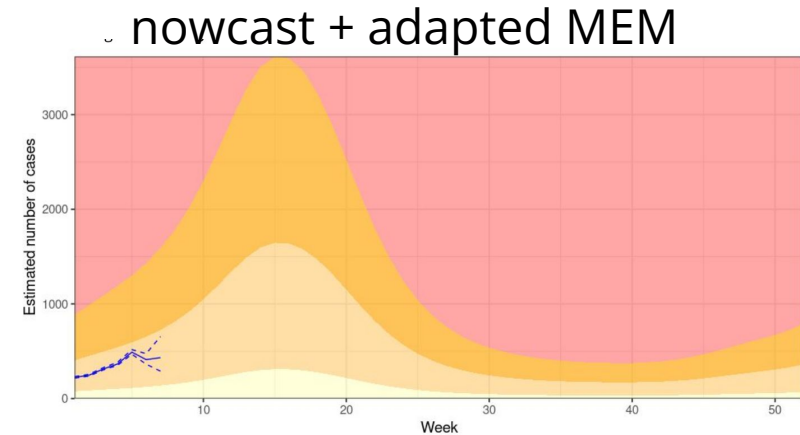
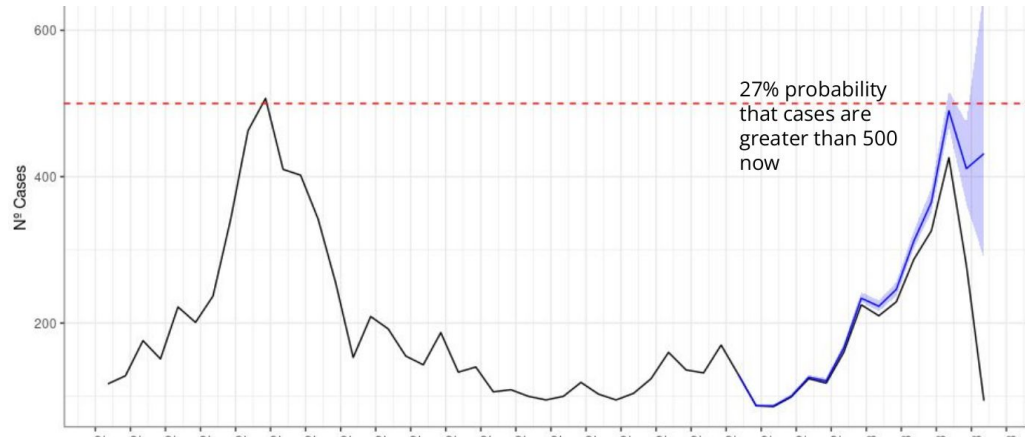
WILEY *Statistic
in Medicine*

A modelling approach for correcting reporting delays in disease surveillance data

Leonardo S Bastos¹  | Theodoros Economou² | Marcelo F C Gomes¹ |
Daniel A M Villela¹ | Flavio C Coelho³ | Oswaldo G Cruz¹ | Oliver Stoner² |
Trevor Bailey² | Claudia T Codeço¹

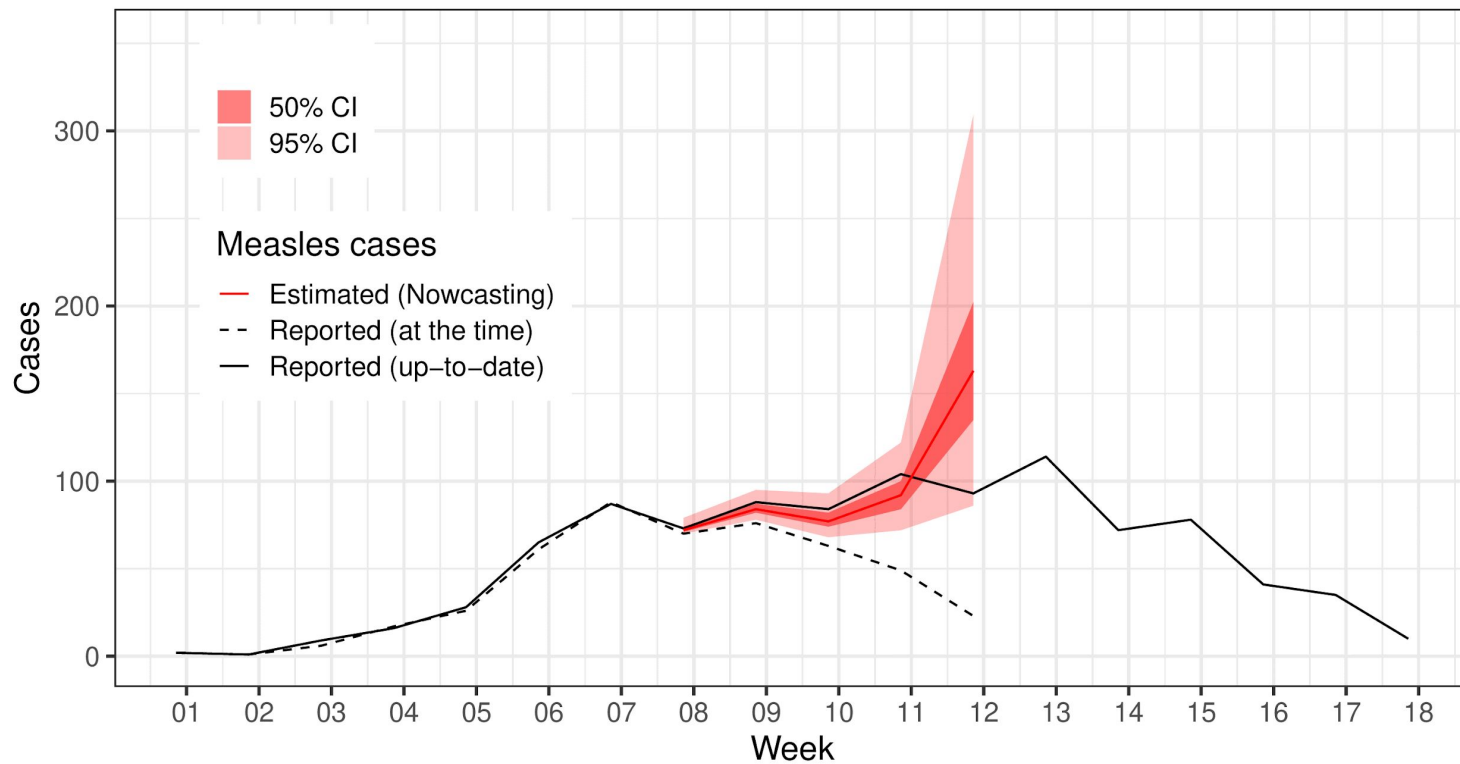


<https://covid19br.github.io/nowcaster/>



Bastos, Gomes & Carvalho (2021)

Measles cases in the US, 2025



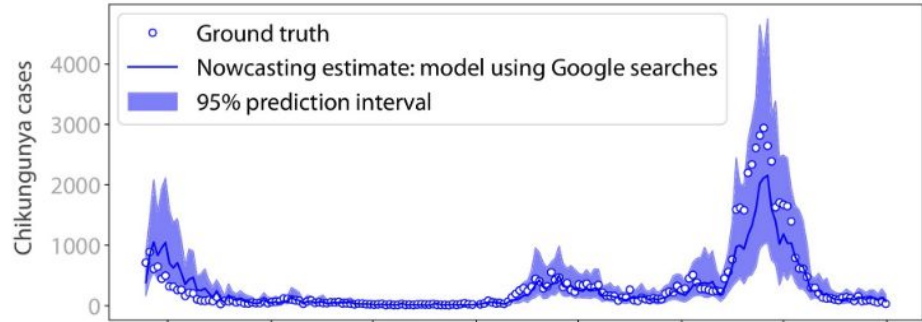
Source: <https://www.cdc.gov/measles/data-research/>

Nowcasting can be improved by online data

RESEARCH ARTICLE

Faster indicators of chikungunya incidence using Google searches

Sam Miller^{1,2*}, Tobias Preis^{1,2}, Giovanni Mizzi¹, Leonardo Soares Bastos³, Marcelo Ferreira da Costa Gomes³, Flávio Codeço Coelho^{4,5}, Claudia Torres Codeço³, Helen Susannah Moat^{1,2*}



Lessons learned:

- nowcasting works if the notification process is stable
 - works better in endemic areas
- accuracy can be improved by including covariates (online data)
- the same method can be used for forecast, but uncertainty is high
 - can be easily extended to other diseases

Infodengue monitors transmission

Epidemics 25 (2018) 101–111



Contents lists available at ScienceDirect

Epidemics

journal homepage: www.elsevier.com/locate/epidemics



Estimating the effective reproduction number of dengue considering temperature-dependent generation intervals

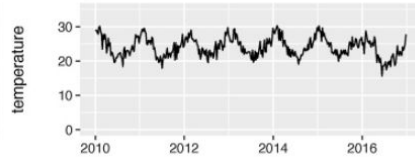


Claudia T. Codeço^{a,1,*}, Daniel A.M. Villela^{a,1}, Flavio C. Coelho^b

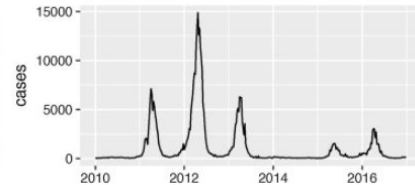
^aScientific Computing Program/Oswaldo Cruz Foundation, Rio de Janeiro, Brazil

^bSchool of Applied Mathematics/Genulio Varas Foundation, Rio de Janeiro, Brazil

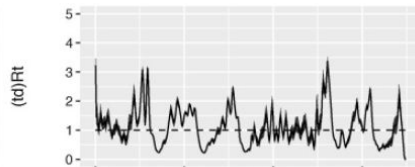
(B) Rio de Janeiro



temperature

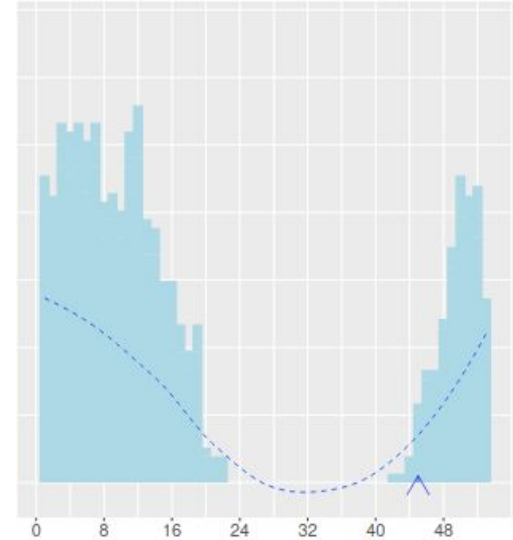


cases

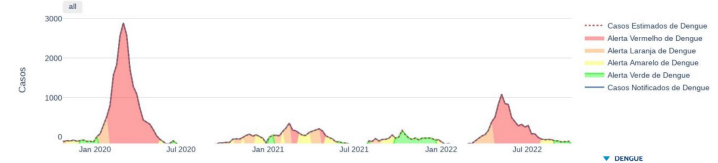


temperature
dependent R_t

transmission seasonality



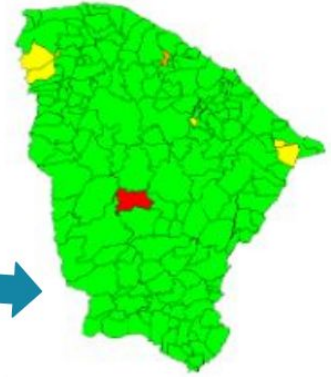
transmission events



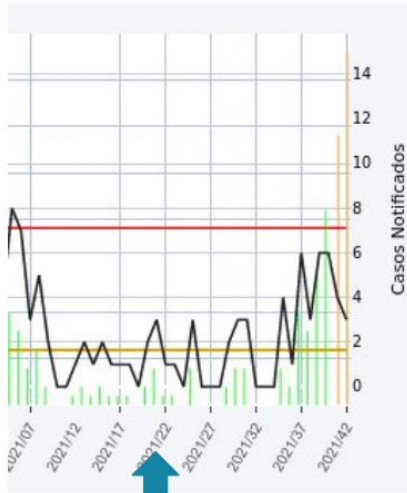
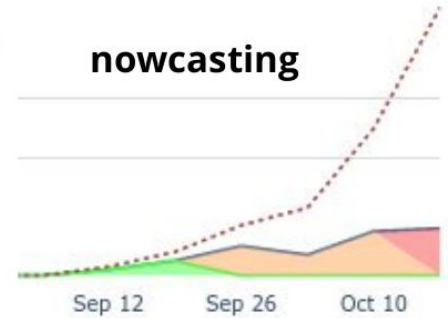
Reports at different spatial scales



Dengue



nowcasting

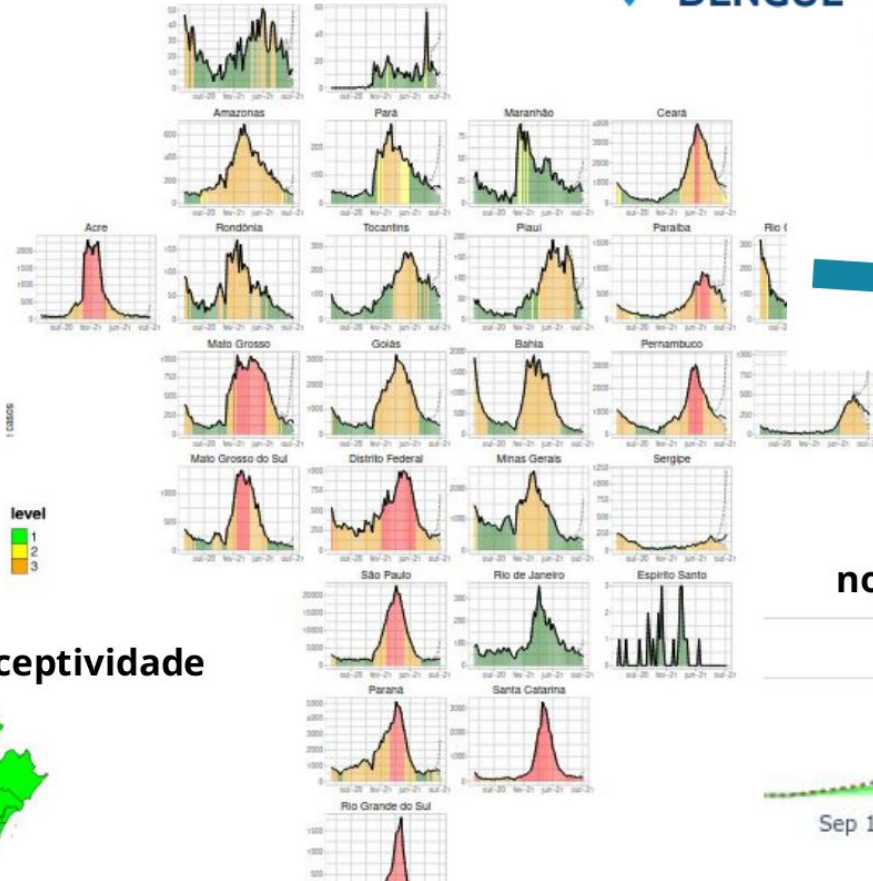


Dengue



receptividade

casos



Fast expansion of dengue in Brazil

Claudia T. Codeco,^{a,*} Sara S. Oliveira,^a Danielle A.C. Ferreira,^b Thais I.S. Riback,^a Leonardo S. Bastos,^a Raquel M. Lana,^{a,c} Iasmim F. Almeida,^a Vinicius B. Godinho,^a Oswaldo G. Cruz,^a and Flavio C. Coelho^d

www.thelancet.com Vol 12 Month August, 2022

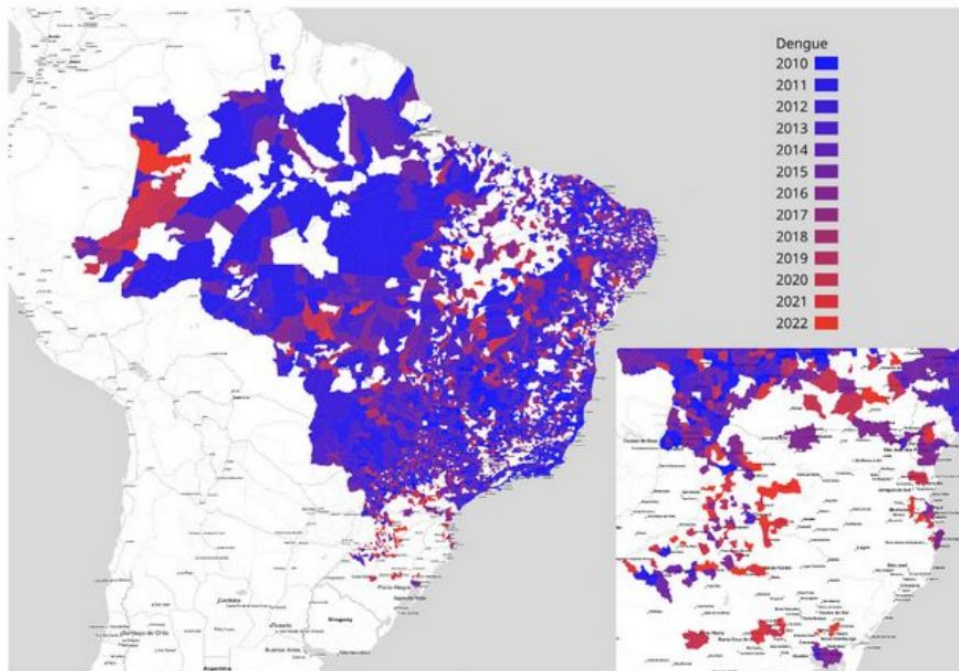


Figure 1. Map of Brazil showing the spread of dengue transmission into previously disease free regions (pre-2010 to March 2022). Source: Infodengue.

Increasing arbovirus risk in Chile and neighboring countries in the Southern Cone of South America

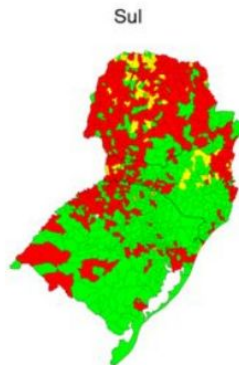
Elizabet L. Estallo,^{a,b,*} Rachel Sippy,^c Michael A. Robert,^d Salvador Ayala,^e Carlos J. Barboza Pizarro,^f Pastor E. Pérez-Estigarribia,^g and Anna M. Stewart-Ibarra^h

Argentina enfrenta riesgo creciente de epidemia de dengue

Doença é conhecida também no país como "febre quebra-ossos"

JUAN BUSTAMANTE E LUCILA SIGAL - REPÓRTERES DA REUTERS

Publicado em 08/04/2024 - 13:31
Buenos Aires



2024

In 2023, Infodengue launched a warning:

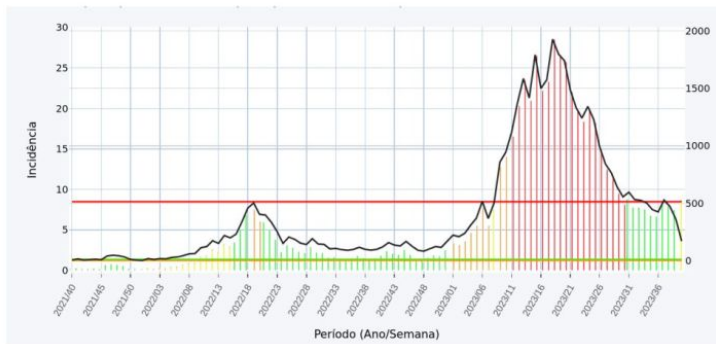
Relatório 02/23 (atualizado em 26/10/2023)

Reflexões sobre o risco de arboviroses em 2024

"The Earth is in Uncharted Territory"

Paulo Ceppi, Imperial College London climate science lecturer, 2023

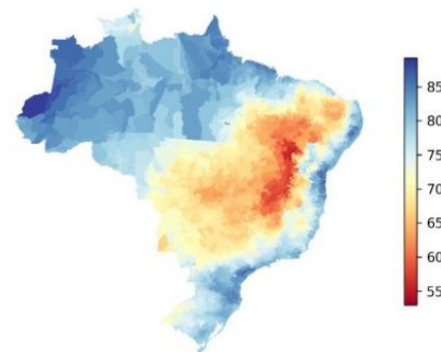
Winter with high notification + El Niño effects = high risk of epidemic



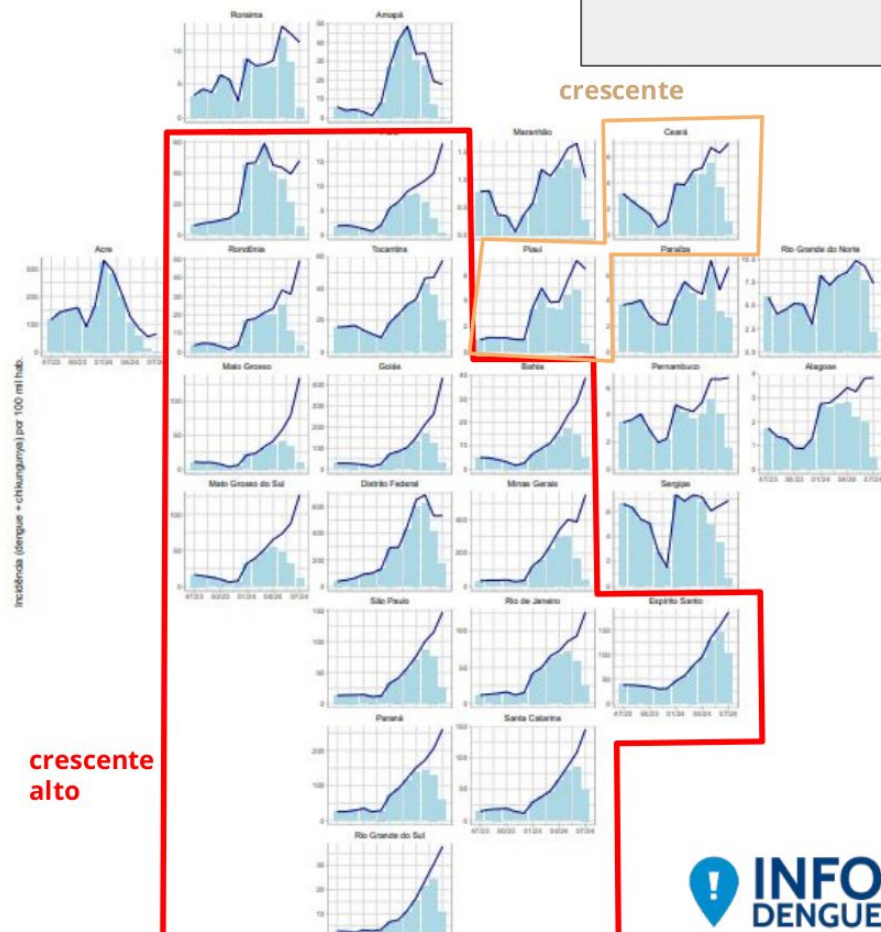
Temperatura média - 2023



Umidade média - 2023



Weekly reports for decision making during 2024



Região	UF	SE limiar alto	SE pico	nível atual SE07	tendência	Rt
Norte	Acre	46	2	médio	decrecente	<1
	Rondônia	3		alto	crescente	1,5
	Roraima			médio	estável	~1
	Amazonas	52	platô	alto	estável	~1
	Pará	2		alto	crescente	1,4
	Amapá	1	3	alto	decrecente	<1
	Tocantins	7		alto	crescente	1,4
Centro-Oeste	Mato Grosso	3		alto	crescente	1,5
	Mato Grosso do Sul	7		alto	crescente	1,6
	Goiás	3		alto	crescente	1,8
	Distrito Federal	49	5	alto	decrecente	~1
Sul	Rio Grande do Sul	1		alto	crescente	1,8
	Santa Catarina	1		alto	crescente	1,7
	Paraná	1		alto	crescente	1,5
Nordeste	Alagoas			baixo	estável	~1
	Bahia	4		alto	crescente	1,8
	Ceará			baixo	crescente	1,3
	Maranhão			médio	estável	~1
	Paraíba			baixo	estável	~1
	Pernambuco			médio	estável	~1
	Piauí			médio	crescente	1,2
	Rio Grande do Norte			médio	estável	~1
	Sergipe			médio	estável	~1
Sudeste	Rio de Janeiro	47		alto	crescente	1,4
	Minas Gerais	1		alto	crescente	1,6
	Espírito Santo	2		alto	crescente	1,6
	São Paulo	3		alto	crescente	1,6

According to the Brazilian MoH, models were useful for:

Preparedness & response

- revise guides
- resources allocation
- launch alerts at local level
- implement and inform the national situation room

Lessons learned in 2024:

- action plans are as important as contingency plans
- little time to act
- dengue is an umbrella for other arboviruses



MoH team

Infodengue Forecast Sprint, 2024

Organization:



Funding:



Teams:

D-fense: Américo cunha (UERJ)

Dobby Data: Eduardo Araújo (FGV)

GeoHealth: Paula Moraga (KAUST)

Global Health Resilience: Rachel Lowe (BSC)

PET: Leo Bastos (PROCC-Fiocruz)

Ki-Dengu Peppa: Luiz Max Carvalho (FGV)

DS_OKSTATE: Lucas Stolerman (OSU)

Main goal: provide forecasts for 2025, at state level, by:

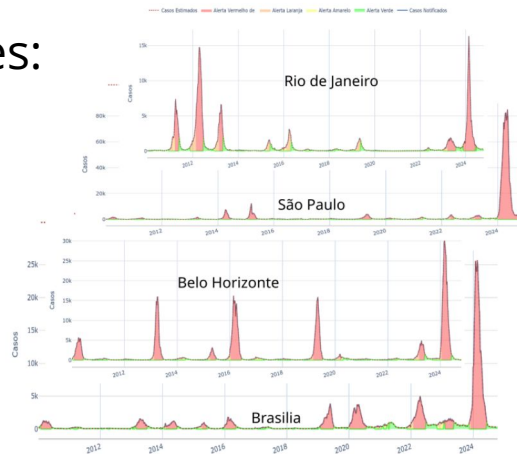
- 1 Organizing a community of modellers with unified goal and methods
- 2 Together generate a set of independent models tested using data from previous seasons
- 3 Train ensemble model with all submissions
- 4 Produce forecasts for 2025 using the best models, either single or combined.
- 5 Technical Report with the results
- 6 Update and monitor the performance of the models until 2025
- 7 Organize a larger Sprint initiative in 2025.

Mosqlimate's Forecast Comparison Platform

The Mosqlimate platform offers data and visualization tools for tracking arbovirus forecasting experiments:

- Open datasets (climatic, demographic and case data):
<https://api.mosqlimate.org/datastore/>
- Model Registry: <https://api.mosqlimate.org/models/>
- Visualization tools:
<https://api.mosqlimate.org/vis/dashboard>
- Forecast scoring tools:
<https://mosqlimate.org/mosqlimate-client/>

Challenges:



Ranking

Full curve challenge

SCOPE	ALL STATES		AM		CE		GO		PR		MG	
TEST YEAR	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024
GOLD	BBM	GHR	BBM	OK	DD	BBM	BBM	GHR	BBM	GHR	BBM	GHR
SILVER	DD	GH	GHR	GH	BBM	GHR	KDP1	GH	KDP1	GH	DD	DD
BRONZE	KDP1	BBM	DD	GHR	GHR	GH	GHR	KDP2	DD	KDP2	KDP1	GHR

Half curve challenge

SCOPE	ALL STATES		AM		CE		GO		PR		MG	
TEST YEAR	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024
GOLD	BBM	GH	BBM	OK	DD	GH	BBM	GHR	BBM	GHR	BBM	GH
SILVER	DD	GHR	DD	GH	BBM	BBM	DD	GH	DD	GH	GHR	KDP2
BRONZE	GHR	OK	GHR	KDP1	GHR	GHR	GHR	DD	KDP1	KDP2	DD	GHR

Peak size

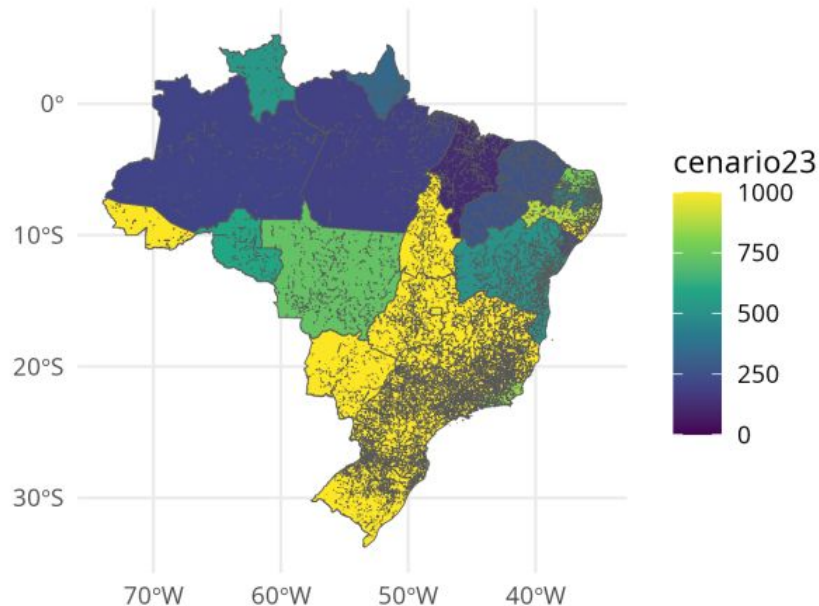
SCOPE	ALL STATES	
TEST YEAR	2023	2024
GOLD	DD	GHR
SILVER	KDP1	GH
BRONZE	BBM	KDP2

Honorable Mentions

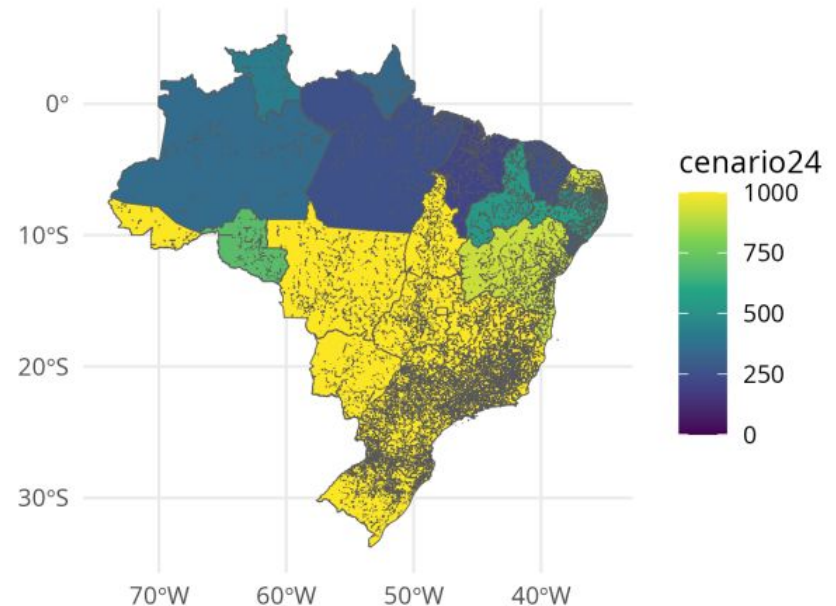
DD predicted peak increase in most states
GHR & KDP accurately predicted the increase in 2024 in MG

Predicted median Incidence maps for 2025

ensemble 1



ensemble 2





2nd Sprint Infodengue – Mosqlimate Dengue Fever Predictive Modeling in Brazil (IMDC)

Registration: May 15, 2025 - June 15, 2025

The **Mosqlimate** and **InfoDengue** teams are hosting the Dengue Fever Predictive Modeling in Brazil (IMDC) for the second year, aiming to promote the development and comparison of predictive models to support epidemiological surveillance in Brazil. Join us!

Register at: <https://sprint.mosqlimate.org/>



Organization



Team



Mosqlimate

Mosqlimate : <http://mosqlimate.org/>
Infodengue: <https://info.dengue.mat.br/>

Fiocruz: Claudia Codeço, Leonardo Bastos, Oswaldo Cruz, Sara Souza,
Thais Riback, Raquel Lana, Iasmim Almeida, Marcelle Chagas,
Lais Freitas, Lucas Bianchi

EMap/FGV: Flavio Coelho, Luã Bida, Eduardo Araujo, Fernanda
Valente, Ezequiel Braga, Julie Souza, Beatriz Laiate

Obrigado!

Support:



MINISTÉRIO DA
SAÚDE



FGV EMap
ESCOLA DE
MATEMÁTICA
APLICADA



IDExtremes